



(To be completed ONLY upon acceptance of supervision by supervisor)

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Project title:

Uncertainty-Conditioned Reinforcement Learning with Evidential Policy Networks for
Autonomous Systems

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By signing this form, the above-mentioned supervisor/s affirms his/her/their intention to supervise the above-mentioned student. The student confirms that in the case of acceptance by the Board of Studies, his/her decision to be supervised by the indicated supervisor is final.

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1 Research Problem and Motivation

Autonomous systems rely on three interconnected components: perception to interpret surroundings, planning to identify optimal routes, and control to execute operations [1]. However, intrinsic uncertainties from sensor noise and inadequate training information propagate across this pipeline [2]. Consequently, existing reinforcement learning approaches rarely combine explicit localisation uncertainty as policy input with calibrated uncertainty estimates in policy outputs. For safe deployment, control policies must react appropriately to these uncertainties.

Examining the literature reveals gaps in current approaches. Most research employs evidential techniques to critic networks [3,4] or applies evidential methods to perception systems for model predictive control [5] rather than policy actors. Other work conditions policies on localisation uncertainty without evidential outputs [6] or uses uncertainty for active exploration [7] rather than adaptive control. Therefore, merging uncertainty as explicit state input with evidential neural network policy outputs presents an unexplored research direction.

2 Proposed Approach

This research proposes an architecture combining two complementary uncertainty handling methods. The technical implementation develops an evidential Proximal Policy Optimisation algorithm [8] in which the actor network outputs Normal Inverse Gamma distributions over continuous actions [9,10], enabling single pass quantification of both epistemic and aleatoric uncertainty. Despite advantages for real time decision making, evidential methods have primarily been demonstrated in perception tasks [11] rather than learned control policies. In contrast, ensemble based methods require multiple forward passes [12].

Simultaneously, an Extended Kalman Filter localisation pipeline extracts covariance matrices within CARLA [13,14] as explicit state input to the policy. Recent work has improved learned covariance estimation for visual odometry [15,16], demonstrating that uncertainty estimates can benefit decision making. However, explicit integration into learned control policies remains unexplored. This dual approach enables control strategies to modify behaviour based on both internal model uncertainty and external localisation confidence.

To achieve this aim, the following objectives have been set:

1. Implement an EKF based localisation pipeline in CARLA to extract covariance matrices as explicit uncertainty features.
2. Develop an evidential PPO algorithm with actor networks outputting NIG distributions over continuous actions.

3. Conduct comparative studies against uncertainty conditioned baselines, evidential critic methods, and standard PPO under varying localisation quality.

3 Evaluation Strategy

Objective 1: The EKF pipeline will be evaluated by comparing estimated poses against CARLA ground truth. Root mean square error will be computed under nominal conditions, then sensor degradation will be introduced to assess covariance quality.

Objective 2: The evidential PPO algorithm will be evaluated on trajectory following tasks including parking manoeuvres. Path tracking error and Expected Calibration Error will be measured.

Objective 3: Comparative studies will be conducted against GUIDE like, EPPO like, and standard PPO baselines under degraded localisation. Safety will be measured through collision rate [17], performance via goal completion rate, and calibration via Expected Calibration Error. The hypothesis is that the proposed architecture could demonstrate improved safety through better calibrated uncertainty estimates.

4 Expected Contributions

This research addresses the perception to control gap in uncertainty quantification. The main contributions include using covariance matrices as explicit state inputs and applying evidential networks to continuous control policies. If verified, this could aid the creation of uncertainty aware autonomous systems capable of conveying confidence levels and applying risk aware decision making.

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